

WILL C. FORTE

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Project Website

EDUCATION

Rutgers University–New Brunswick, New Brunswick, NJ

Projected Graduation: May 2027

B.S. in Mechanical Engineering with Minor in CS

- Aresty Fellow (1.96% acceptance) | Engineering Honors Academy Scholar (top 5.7% of incoming class) | 3.78 GPA

Relevant Coursework: Honors Intro Linear Algebra, Honors Calculus III, Honors Statics, CAD, Leadership Communication

RESEARCH EXPERIENCE

Aresty Research Fellow, **Rutgers PRACSYS Lab** (PI: Kostas Bekris), New Brunswick, NJ

March 2025 - Present

- Selected as a robotics researcher (1.96% acceptance rate) to research neural networks for teleoperation
- Trained diffusion and flow-matching models for imitation learning using Meta FAIR libraries
- Invited with funding to attend the ICRA 2025 Undergrad Outreach Workshop in Atlanta, GA

Research Assistant, **Rutgers RAM Lab**, Piscataway, NJ

October 2024 - March 2025

- Implemented scikit-learn ML to approximate the regions of attraction of dynamical systems
- Designed the first all-aluminum chassis for Rutgers' two-legged robot to reduce model uncertainty
- Reproduced CMU Lidar-based ROS SLAM stack for autonomous quadruped exploration

Research Assistant, **Rutgers Advanced Controls Lab**, Piscataway, NJ

June 2024 - October 2024

- Developed low-level quadcopter control architecture in ROS for PX4-MAVROS-Gazebo SITL simulator
- Created a C++ PX4 library to control positional servos on a tiltrotor quadcopter

Research Assistant, **NJIT Swissler Innovative Robotics Lab**, Newark, NJ

June 2023 - June 2024

- Developed a prismatic robotic arm simulation in MuJoCo (Python) using Jacobian inverse kinematics
- Programmed ESP32 microcontrollers (C++) with onboard computer vision (OpenCV) in ESP-IDF

PUBLICATIONS

N. Vijay, **W. C. Forte**, I. Gajjar, S. Patham, S. Gupta, S. Shah, P. Trivedi, and R. Arora, "Acoustic analysis of uneven blade spacing and toroidal geometry for reducing propeller annoyance," arXiv preprint arXiv:2504.12554, Apr. 16, 2025. <https://arxiv.org/abs/2504.12554>

PROJECTS

Fully-Actuated Quadrupedal Robot ([Webpage on willcforte.com](http://willcforte.com))

- Designed and fabricated a 12-servo quadruped robot using spare FTC motors, 3D-printing, and laser-cut gears
- Created model-based PID control loop using MuJoCo Python bindings and embedded system to execute a given pose

Cuff-Link Electromyographic Assistive Device ([Webpage on willcforte.com](http://willcforte.com))

- Developed assistive device for amputees to control PC via arm muscles, outperforming trackpad aim by 126 ms
- Presented final product to a crowd of 200+ at the 2024 Academy Engineering Showcase (2nd place)

Toroidal and Uneven Blade Aeroacoustic Analysis (TUBAA; Website at tubaa.dev)

- Conducted CFD analysis of MIT Lincoln Lab toroidal propellers in ANSYS Fluent, showing reduction in tip vortices
- Mentored by Stanford PhD Candidate Gao Jun Wu and UC Davis Professor Seongkyu Lee

AWARDS

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| • Rutgers Aresty Fellowship (1.96% acceptance rate) | 2025 |
| • Air Force Research Lab Philips Scholar (declined 2x) | 2025 |
| • Rutgers Engineering Honors Academy Scholar (top 65 out of 1140 incoming engineers) | 2024 |
| • Yale Physics Olympics Fermi Estimation 2nd Place | 2023 |
| • FTC Robotics Think Award | 2023 |
| • NJAPT Physics Olympics Champions (first in school's history – 15 years) | 2023 |

LEADERSHIP & EXTRACURRICULARS

Research & Development Team Lead, RUAutonomous, Rutgers AIAA April 2025 - Present

- Forming a hardware and software-focused subteam to test new drone configurations, notably VTOL & coaxial motors

Event Photographer, Rutgers Honors College Media Team March 2025 - Present

TALKS

Workshop: Introduction to the MuJoCo Simulator, Rutgers University, N2E Robotics Club, February 2025

The Cuff-Link Assistive Device, Academy for Math, Sci., & Engg., AMSE Senior Showcase, May 2024

PROFESSIONAL DEVELOPMENT

L'SPACE NPWEE Academy, **National Aeronautics and Space Administration (NASA)** May 2025 - Present

- Selected by NASA Marshall to write a space robotics proposal for the Lucy mission

Invited Attendee, ICRA 2025 Undergraduate Outreach Workshop, Atlanta, GA May 2025

SKILLS

Software: ROS/ROS2, Ubuntu, Git, OpenCV, MuJoCo, Isaac Sim, Docker, Gazebo, PX4, ESP32, Arduino

ML/AI: PyTorch, FAIR flow_matching, Slurm

Languages: C++, Python, MATLAB, Java, $\text{\LaTeX}$, Shell Scripting, CMake, Vue.js, Node.js, HTML/CSS/JS, Flutter

CAD/CFD: SOLIDWORKS, Fusion360, Onshape, ANSYS Fluent, AutoCAD

Fabrication: CAM/CNC, Manual Mill, 3D Printing, Laser Cutting, Plasma Cutting, Soldering, Electrical Schematics